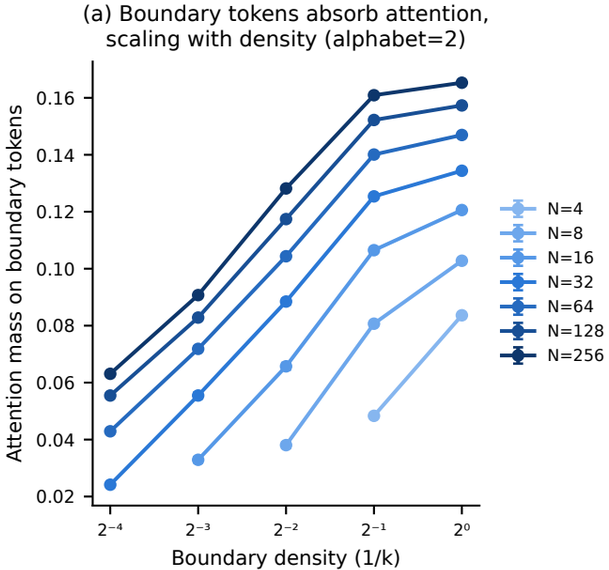
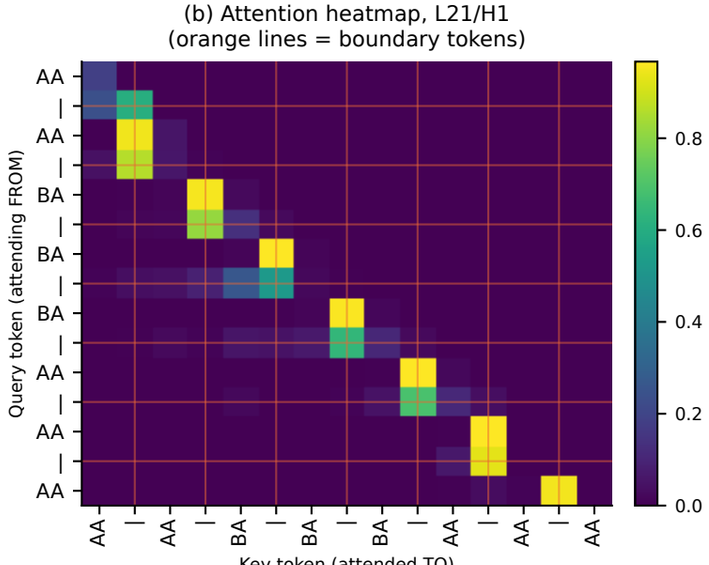
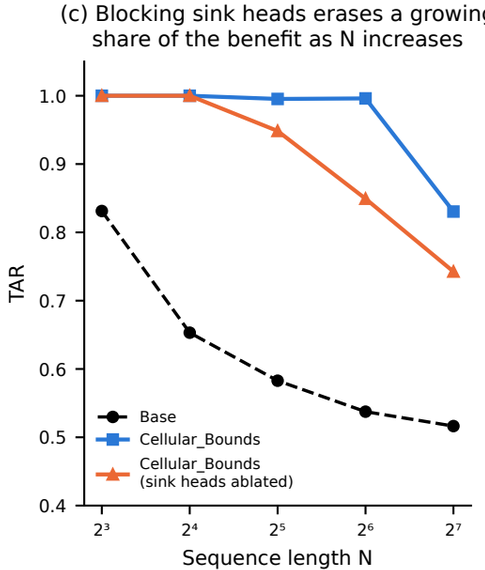
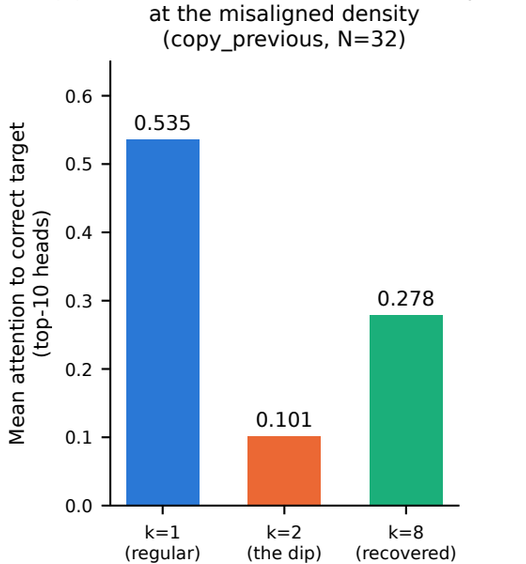

(a) Boundary tokens absorb attention, scaling with density (alphabet=2)
Attention mass on boundary tokens
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
N=4
N=8
N=16
N=32
N=64
N=128
N=256
2⁻⁴
2⁻³
2⁻²
2⁻¹
2⁰
Boundary density (1/k)


(b) Attention heatmap, L21/H1 (orange lines = boundary tokens)
Query token (attending FROM)
AA
|
AA
|
BA
|
BA
|
BA
|
AA
|
AA
|
AA
AA
|
AA
BA
|
BA
|
BA
|
AA
|
AA
|
AA
Key token (attended TO)
0.8
0.6
0.4
0.2
0.0


(c) Blocking sink heads erases a growing share of the benefit as N increases
TAR
1.0
0.9
0.8
0.7
0.6
0.5
0.4
Base
Cellular_Bounds
Cellular_Bounds
(sink heads ablated)
2³
2⁴
2⁵
2⁶
2⁷
Sequence length N


(d) Attention to the needed token collapses at the misaligned density (copy_previous, N=32)
Mean attention to correct target (top-10 heads)
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0.535
0.101
0.278
k=1
(regular)
k=2
(the dip)
k=8
(recovered)